

Experiment No. – 4

Aim –

- (A.) You are given a table `tbl_happiness` that contains the ranking of happiness for different countries. Your task is to retrieve all the records from the table **while ensuring that "India" and "Sri Lanka" appear at the top, followed by the rest of the countries in their original ranking order.**
- (B.) Make all the rankings in descending order except **India** and **Sri Lanka**
- (C.) A company maintains employee information in a single table where each employee may report to another employee acting as their manager. Prepare a report that displays the details of every employee who supervises at least one other employee. The report should include the manager's employee ID, name, total number of direct reports, and the average age of those reporting employees. The average age should be rounded to the nearest whole number and the final result should be sorted by employee ID.

Tools Required

- Postgres
- PgAdmin4

Queries:

A.

```
CREATE TABLE tbl_happiness
(
    sno INT PRIMARY KEY,
    rankings INT,
    country VARCHAR(50)
);

INSERT INTO tbl_happiness
VALUES
(1,1,'Finland'),
(2,2,'Denmark'),
(3,3,'Iceland'),
(4,4,'Israel'),
```

```
(5,5,'Netherlands'),  
(6,6,'Sweden'),  
(7,7,'Norway'),  
(8,126,'India'),  
(9,128,'Sri Lanka');
```

```
SELECT *,  
    CASE  
        WHEN country = 'India' THEN 1  
        WHEN country = 'Sri Lanka' THEN 2  
        ELSE 3  
    END AS DERIVED_RANK  
FROM tbl_happiness  
ORDER BY DERIVED_RANK, RANKINGS DESC; (4, 'D2', 8.00),
```

B.

```
SELECT *  
FROM tbl_happiness  
ORDER BY  
CASE  
    WHEN country = 'India' THEN 1  
    WHEN country = 'Sri Lanka' THEN 2  
    ELSE 3  
END,  
CASE  
    WHEN country IN ('India', 'Sri Lanka') THEN rankings  
    ELSE -rankings  
END;
```

C.

```
CREATE TABLE Employees(  
    employee_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    reports_to INT,  
    age INT  
);  
  
INSERT INTO Employees (employee_id, name, reports_to, age)  
VALUES  
(9, 'Hercy', NULL, 43),  
(6, 'Alice', 9, 41),  
(4, 'Bob', 9, 36),  
(2, 'Winston', NULL, 37);  
  
SELECT  
    m.employee_id,  
    m.name,  
    COUNT(e.employee_id) AS reports_count,  
    ROUND(AVG(e.age), 0) AS average_age  
FROM Employees AS m  
JOIN Employees AS e  
ON m.employee_id = e.reports_to  
GROUP BY  
    m.employee_id,  
    m.name  
ORDER BY  
    m.employee_id;
```

Outputs:

	sno [PK] integer	rankings integer	country character varying (50)	derived_rank integer
1	8	126	India	1
2	9	128	Sri Lanka	2
3	7	7	Norway	3
4	6	6	Sweden	3
5	5	5	Netherlands	3
6	4	4	Israel	3
7	3	3	Iceland	3
8	2	2	Denmark	3
9	1	1	Finland	3

Showing rows: 1 to 9



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	sno [PK] integer	rankings integer	country character varying (50)
1	8	126	India
2	9	128	Sri Lanka
3	7	7	Norway
4	6	6	Sweden
5	5	5	Netherlands
6	4	4	Israel
7	3	3	Iceland
8	2	2	Denmark
9	1	1	Finland

Learning Outcome

- Use the CASE statement within ORDER BY to implement custom sorting logic.
- Prioritize specific records while maintaining the order of other records.
- Apply ascending and descending sorting selectively based on conditions.
- Solve practical SQL ordering problems using conditional expressions.